

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, rev = 0, remainder;
6     int count = 0;
7
8     scanf("%d", &n);
9
10    while (n != 0)
11    {
12        count++;
13        remainder = n % 10;
14        count++;
15
16        rev = rev * 10 + remainder;
17        count++;
18        n = n / 10;
19        count++;
20    }
21
22    count += 3;
23    printf("%d", count);
24
25    return 0;
26 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓